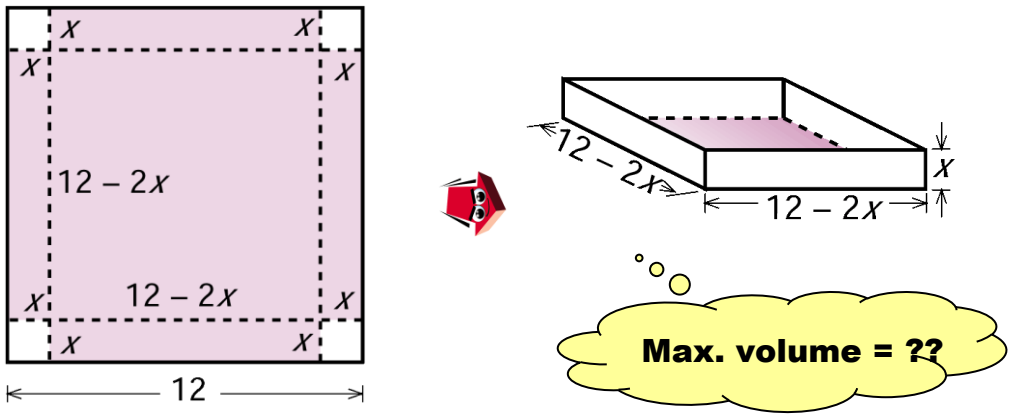
Optimum Problems (最優化問題)

在日常生活中，我們會用到微分 (Differentiation) 去解決一些問題 :

e.g. 現在我們有一張邊長為 12 cm 的正方形紙，如果想用它來製作一個沒有蓋的盒子 (如下圖)，求可製成的最大體積 (maximum volume).

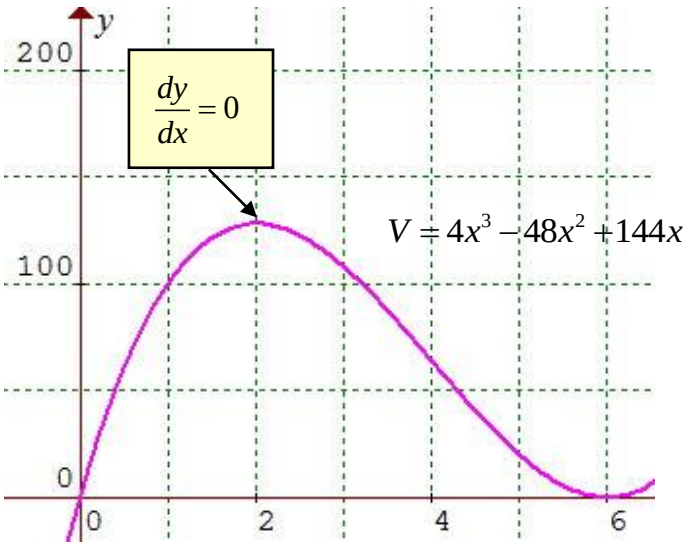


首先我們設盒子的體積為 V cm^3，接下來

$$V = x(12-2x)^2 \quad \text{其中} \quad 0 < x < 6$$

$$V = 4x^3 - 48x^2 + 144x$$

以下是 $V = 4x^3 - 48x^2 + 144x$ 的圖像 :



我們可見當 $x = 2$ 時，

V 的值是最大的!

所以盒子最大的體積是

$V = 2[12-2(2)]^2 = 128\,\text{cm}^3$

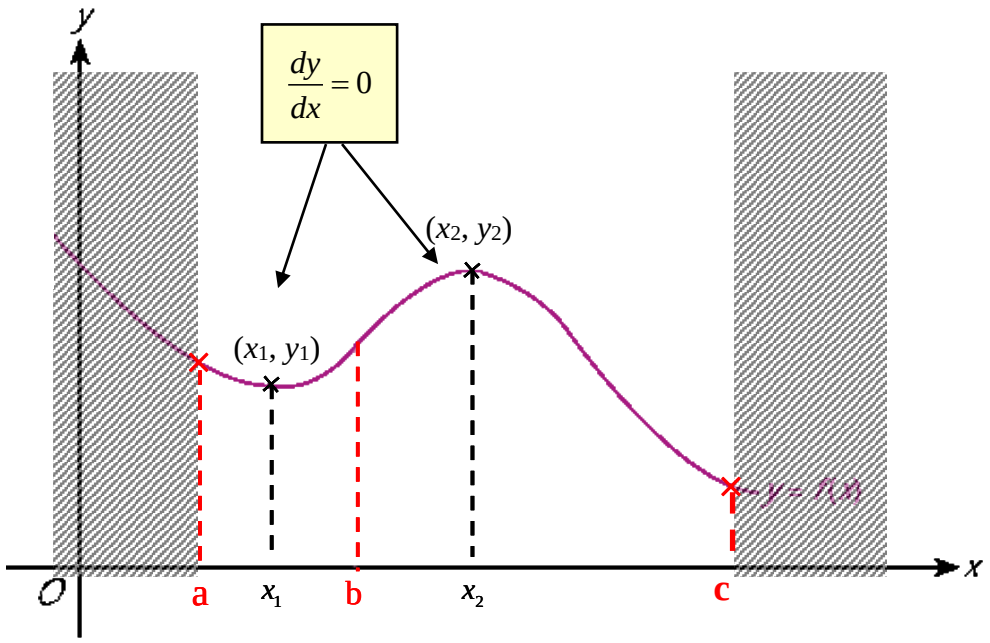
除了繪圖外，我們有更快捷的方法去完成上述步驟。

Not for sale

Optimum Problems

Optimum Problems 最優化問題（求最大或最小值）:

還記得怎樣去求曲線的最大值 / 最小值嗎?



我們可見全局最大點 / 最小點會出現於以下位置 :

$x = x_1$ **(local min),**	$x = x_2$ **(local max),**	$x = a$ **or** $x = c$ **(end points).**

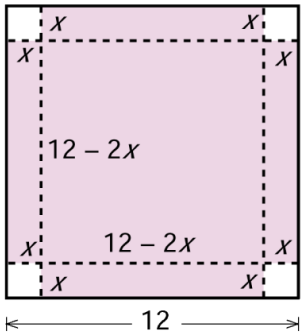
所以我們可以用微分 (Differentiation) 去解決最優化問題 :

Step 1:	將問題以函數式表示及將它微分 (Differentiate).
Step 2:	設 $\frac{dy}{dx} = 0$ 去求出函數的駐點 (Stationary points).
Step 3:	用 **導數判別法 (Derivative Test)** 去求 **相對最大值 / 最小值 (relative max / min values)**.
Bonus step:	求出 **全局最大值 / 最小值 (global max / min values)** .
Step 4:	寫下結論 (Draw the conclusion).

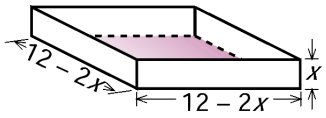
Not for sale

Optimum Problems

e.g. 現在我們有一張邊長為 12 cm 的正方形紙，如果想用它來製作一個沒有蓋的盒子 (如下圖)，求可製成的最大體積 (maximum volume).







首先我們設盒子的體積為 V cm^3，接下來

$V = x(12-2x)^2$ 其中 $0 < x < 6$

$V = 4x^3 - 48x^2 + 144x$

Step 1 - Differentiate the function

$V = 4x^3 - 48x^2 + 144x$

$\dfrac{dV}{dx} = 12x^2 - 96x + 144$

Step 2 - Set the above derivative = 0

Set $\dfrac{dA}{dx} = 0 \Rightarrow$ $V = 12x^2 - 96x + 144$

$x = 2$ or 6 (rej.)

Step 3 - Test of the max. / min. point(s)

$\dfrac{dV}{dx} = 12x^2 - 96x + 144$

$\dfrac{d^2V}{dx^2} = 24x - 96$

when x = 2, $\dfrac{d^2V}{dx^2} < 0$ $\therefore$ V is max when $x = 2$

Step 4 - Conclusion

$\therefore$ Max. volume of $V = 4(2)^3 - 48(2)^2 + 144(2)$ $= 128$ cm^3

一定要檢查所求出的 x 值是否在已知區間內！

在此你可用一階或二階導數判別法。

Not for sale

HKCEE Questions (level 2)

1. ***[A.Math 1986 Paper I Q.12]***

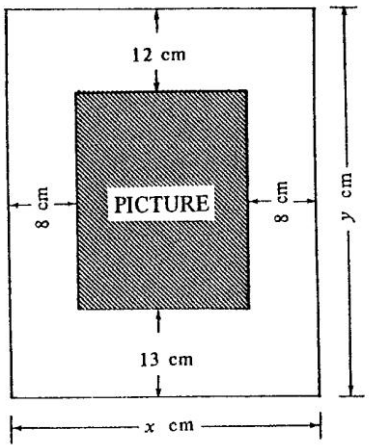


Figure 3

Figure 3 shows a rectangular picture of area A cm^2 mounted on a rectangular piece of cardboard of area 3600 cm^2 with sides of length x cm and y cm. the top, bottom and side margins are 12 cm, 13 cm and 8 cm respectively.

(a) Find A in terms of x. ***(2 marks)***

(b) Show that the largest value of A is 1600. ***(5 marks)***

(c) **(i)** Find the range of value of x for which A decreases as x increases.

(ii) If $x \geq 50$, find the largest value of A.

(6 marks)

Not for sale

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

2. ***[A.Math 1999 Paper I Q.12]***

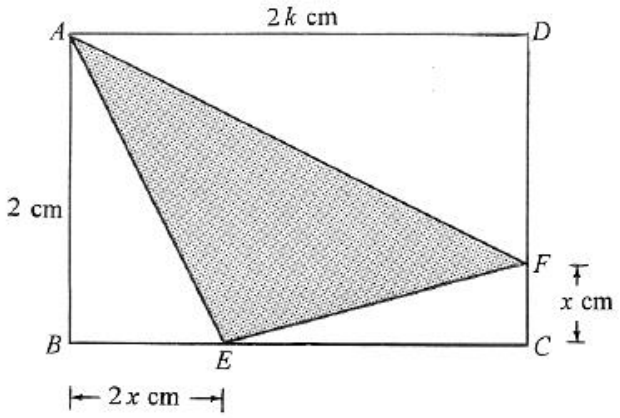


Figure 5

Figure 5 shows a rectangle $ABCD$ with $AB = 2$ cm and $AD = 2k$ cm, where k is a positive number. E and F are two variable points on the sides BC and CD respectively such that $CF = x$ cm and $BE = 2x$ cm, where x is a non-negative number. Let S cm^2 denote the area of ΔAEF.

(a) Show that $S = x^2 - 2x + 2k$. ***(3 marks)***

(b) Suppose $k = \frac{3}{2}$.

(i) By considering that points E and F lie on the sides BC and CD respectively, show that $0 \le x \le \frac{3}{2}$

(ii) Find the least value of S and the corresponding value of x.

(iii) Find the greatest value of S. ***(9 marks)***

(c) Suppose $k = \frac{3}{8}$. A student says that S is least when $x = 1$.

(i) Explain whether the student is correct.

(ii) Find the least value of S. ***(4 marks)***

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

3. ***[A.Math 1993 Paper Q.9]***

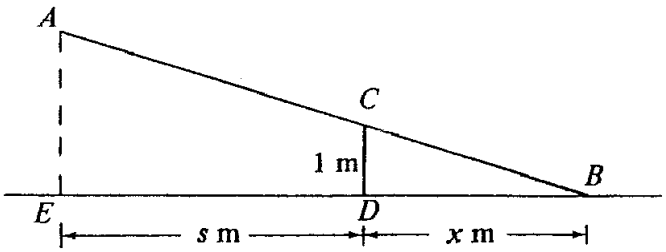


Figure 2

Figure 2 shows a straight rod *AB* of length 8 m resting on a vertical wall *CD* of height 1m.The end *B* is free to slide along a horizontal rail such that *AB* is vertically above the rail. Let *E* be the projection of *A* on rail, *DE* = *s* m and *BD* = *x* m, where $0 < x < 3\sqrt{7}$.

(a) Show that $s = \dfrac{8x}{\sqrt{1+x^2}} - x$. ***(3 marks)***

(b) Find the maximum value of *s*. ***(7 marks)***

(c) Let *P* m^2 be the area of the trapezium *CAED*.

(i) Show that $P = \dfrac{32x}{1+x^2} - \dfrac{x}{2}$

(ii) Does *P* attain a maximum when s attains its maximum? Explain your answer.

(6 marks)

Not for sale

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

4. ***[A.Math 2002 Paper Q.14]***

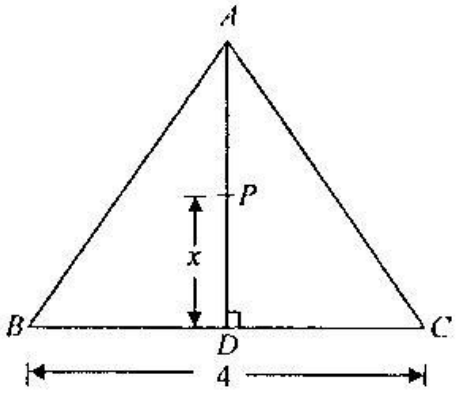


Figure 5

Figure 5 shows an isosceles triangle ABC with $AB = AC$ and $BC = 4$. D is the foot of perpendicular from A to BC and P is a point on AD. Let $PD = x$ and $r = PA + PB + PC$, where $0 \le x \le AD$.

(a) Suppose that $AD = 3$.

(i) Show that $\dfrac{dr}{dx} = \dfrac{2x}{\sqrt{x^2+4}} - 1$.

(ii) Find the range of values of x for which

(1) r is increasing,

(2) r is decreasing.

Hence, or otherwise, find the least value of r.

(iii) Find the greatest value of r. ***(9 marks)***

(b) Suppose that $AD = 1$. Find the least value of r. ***(3 marks)***

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

5. ***[A.Math 1991 Paper Q.11]***

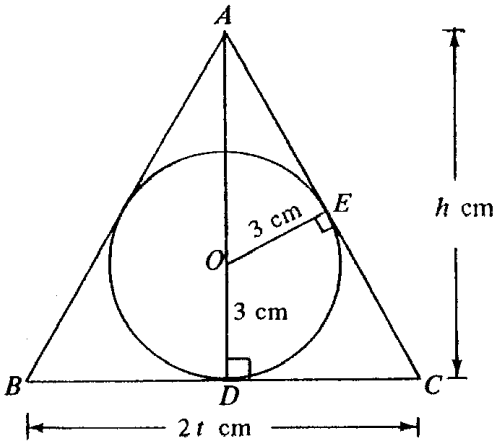


Figure 2(a)

ABC is a variable isosceles triangle with *AB* = *AC* such that the radius of its inscribed circle is 3 cm. the height *AD* and the base *BC* of ΔABC are *h* cm and 2*t* cm respectively, where $h > 6$. (See Figure 2(a).) Let *p* cm be the perimeter of ΔABC.

(a) Show that $t^2 = \dfrac{9h}{h-6}$. ***(3 marks)***

(b) Show that $p = \dfrac{2h^{\frac{3}{2}}}{(h-6)^{\frac{1}{2}}}$. ***(3 marks)***

(c) **(i)** Find the range of values of h for which $\dfrac{dp}{dh}$ is positive.

(ii) Find the minimum value of *p*.

(6 marks)

(d) **(i)** Sketch the graph of *p* against *h* for $h > 6$.

(ii) Hence write down the range of values of *p* for which two different isosceles triangles whose inscribed circles are of radii 3 cm can have the same perimeter *p* cm.

(4 marks)

Not for sale

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 3)

7. ***[A.Math 1981 Paper I Q.9]***

A man is to make a tank of capacity V cubic meters from thin metal sheets. The tank is to consist of a right circular cylinder and two hemispheres, as shown in Figure 1. The cylinder is of length h meters and radius r meters.

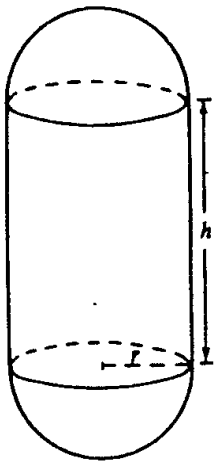


Figure 1

(a) Express h in terms of r and V. ***(3 marks)***

(b) The cost per square meter of the cylindrical surface is k while that of the hemispherical surfaces is $2k$. Let the cost for making the tank be C.

(i) Show that $C = \frac{16}{3}\pi r^2 k + \frac{2kV}{r}$.

(ii) If $\frac{dC}{dr} = 0$, find r in terms of V.

Show that this value of r gives a minimum value of C.

(iii) If C is to be a minimum, find the ratio r : h.

(17 marks)

Not for sale

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

8. ***[A.Math 1988 Paper I Q.12(a)]***

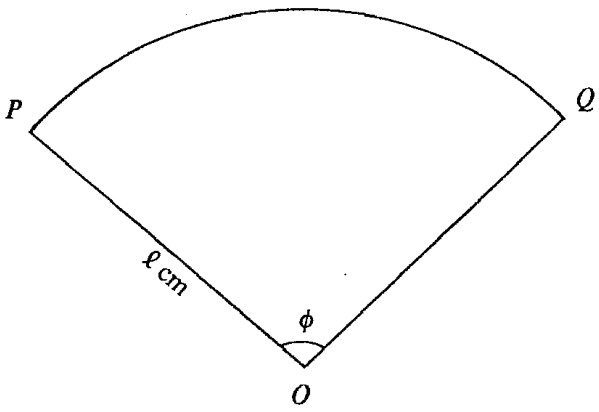


Figure 3(a)

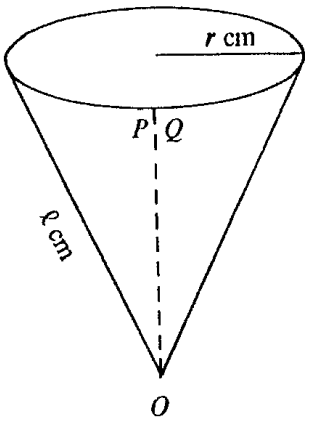


Figure 3(b)

Figure 3(a) shows a piece of paper in the shape of sector of radius l cm and $\angle POQ = \phi$ radians. It is made to form a conical vessel of radius r cm where OP coincides with OQ (see Figure 3(b)).

(i) Express r in terms of l and ϕ.

Let V cm^3 be the capacity of the vessel.

(ii) Show that $V^2 = k(4\pi^2\phi^4 - \phi^6)$ where $k = \dfrac{l^6}{576\pi^4}$.

(iii) By finding $\dfrac{d(V^2)}{d\phi}$, determine the value of ϕ for which the capacity of the vessel is a maximum.

(Note: You may use the fact that V is a maximum when V^2 is a maximum.)

(10 marks)

Not for sale

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

9. ***[A.Math 1997 Paper I Q.12]***

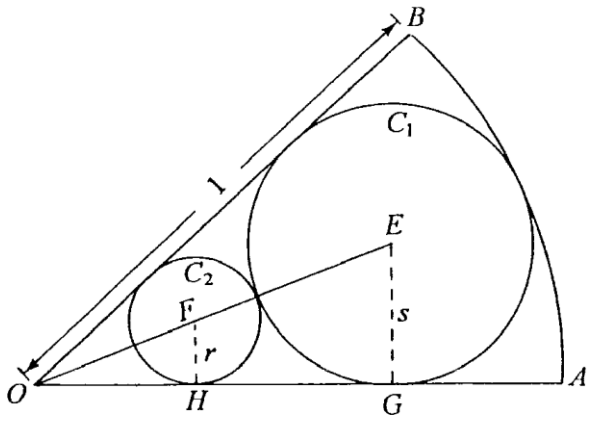


Figure 4

In figure 4, *OAB* is a sector of unit radius and $\angle AOB = 2\theta$, where $0 < \theta < \frac{\pi}{2}$. C_1 is an inscribed circle of radius *s* in the sector. C_2 is another circle of radius *r* touching *OA*, *OB* and C_1. Let *E* and *F* be the centers of C_1 and C_2 respectively. *OA* touches C_1 and C_2 at *G* and *H* respectively.

(a) Show that $s = \frac{\sin\theta}{1+\sin\theta}$. Hence find $\frac{ds}{d\theta}$. ***(4 marks)***

(b) By considering ΔOFH and ΔOEG, express *r* in terms of *s*.

Hence show that $\frac{dr}{d\theta} = \frac{\cos\theta(1-3\sin\theta)}{(1+\sin\theta)^3}$ ***(5 marks)***

(c) By considering the ranges of values of θ for which *r* is

(i) increasing, and

(ii) decreasing,

find the maximum area of circle C_2 [Note: You may give your answers correct to three significant figures.] ***(5 marks)***

(d) Does the area of circle C_1 attain a minimum when the area of circle C_2 attains its maximum? Explain your answer. ***(2 marks)***

Not for sale

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

10. ***[A.Math 2007 Q.16]***

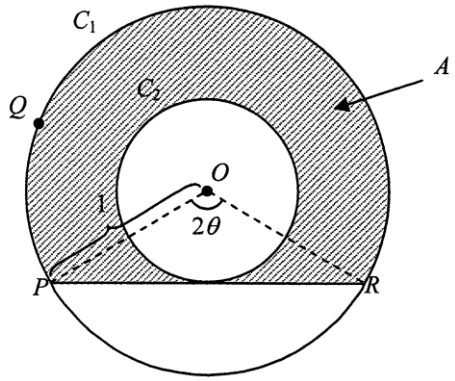


Figure 9

C_1 is a circle with centre O and radius 1. PR is a variable chord which subtends an angle 2θ at O, where $0 < \theta < \frac{\pi}{2}$. C_2 is a circle with centre O and touches PR. Let the area of the shaded region bounded by C_1, C_2 and PR be A (see Figure 9).

(a) Show that

(i) $A = \pi \sin^2 \theta - \theta + \frac{1}{2}\sin 2\theta$

(ii) $\frac{dA}{d\theta} = (\pi - \tan\theta)\sin 2\theta$ ***(5 marks)***

(b) When A attains its greatest value, find the value of $\tan\theta$. ***(3 marks)***

(c) A student guess that when A attains its greatest value, the perimeter of the shaded region will also attain its greatest value. Explain whether the student's guess is correct or not. ***(4 marks)***

[Note: the perimeter of the shaded region = PQR + PR + circumference of C_2.]

Not for sale

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

11. ***[A.Math 2010 Q.13]***

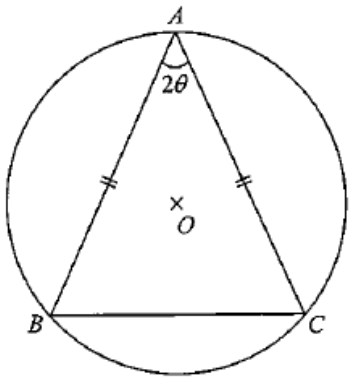


Figure 3

Figure 3 shows a circle with centre *O* and radius 1. A triangle *ABC* is inscribed in the circle with $AB = AC$. Let $\angle BAC = 2\theta$, where $0 < \theta < \frac{\pi}{4}$.

(a) Let *S* be the area of ΔABC.

(i) Show that $S = \frac{\sin 4\theta}{2} + \sin 2\theta$.

(ii) Find the maximum area of ΔABC. ***(7 marks)***

(b)

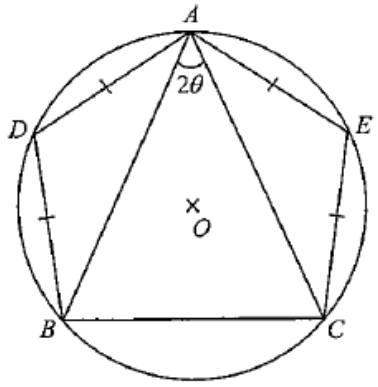


Figure 4

Two points *D* and *E* are added on the circle in Figure 3 such that $AD = BD = AE = CE$ (see Figure 4). When the area of ΔABC attains its maximum, will the area of pentagon *ADBCE* also attain the maximum? Explain your answer. ***(5 marks)***

Not for sale

HKCEE Questions (level 3)

Not for sale

HKCEE Questions (level 3)

12. ***[A.Math 1987 Paper I Q.9]***

(a)

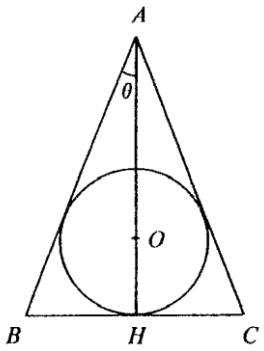


Figure 4(a)

Figure 4(a) shows a circle of center O and radius a inscribed in an isosceles triangle ABC with $AB = AC$. Let $\angle OAB = \theta$.

(i) Find, in terms of a and θ, the height AH of ΔABC.

Hence show that the area of ΔABC is $\dfrac{a^2(1+\sin\theta)^2}{\sin\theta\cos\theta}$.

(ii) For what value of θ is the area of ΔABC a minimum?
(Testing of maximum/minimum is not required.)

(10 marks)

(b)

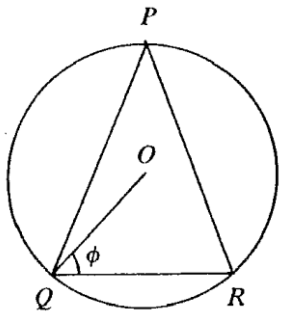


Figure 4(b)

Figure 4(b) shows a circle of center O and radius b circumscribing an isosceles triangle PQR with $PQ = PR$. Let $\angle OQR = \phi$.

(i) Show that the area of ΔPQR is $b^2\cos\phi(1+\sin\phi)$.

(ii) When ΔPQR is equilateral, show that its area is a maximum.

(10 marks)

Not for sale

HKCEE Questions (level 3)

Not for sale

More Questions in Past Papers

1. ***[A.Math 1984 Paper I Q.10]***

In Figure 2, *ABCD* is a square tin plate of side $2\sqrt{2}$ m. *PQRS* is a square whose center coincides with that of *ABCD*. The shaded parts are cut off and the remaining part is folded to form a right pyramid with base *PQRS*. Let $PQ = 2x$ meters and let the volume of the pyramid is V cubic meters.

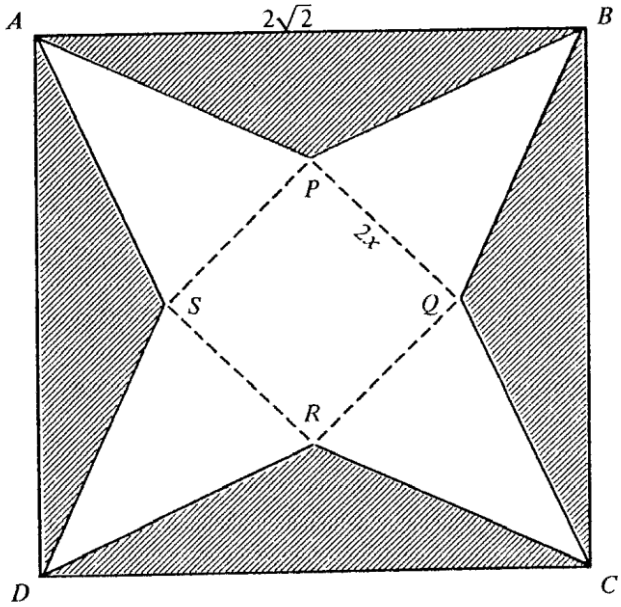


Figure 2

(a) Show that the height of the pyramid is given by $2\sqrt{1-x}$ meters.

Hence express V as a function of x.

(8 marks)

(b) Find the stationary points of the graph of V.

Find the equations of the tangents to the graph at the stationary points and at $x = 1$.

Hence sketch the graph for $0 \le x \le 1$.

(12 marks)

Not for sale

More Questions in Past Papers

2. ***[A.Math 2009 Q.16]***

(a) Let $f(x)=(14-x)(x^2+9)$.

(i) Find the coordinates of all maximum and minimum points of the curve $y=f(x)$.

(ii) Sketch the graph of $y=f(x)$ for $0\le x\le 14$. ***(5 marks)***

(Note: the suggested range of values of y is $0\le y\le 500$)

(b)

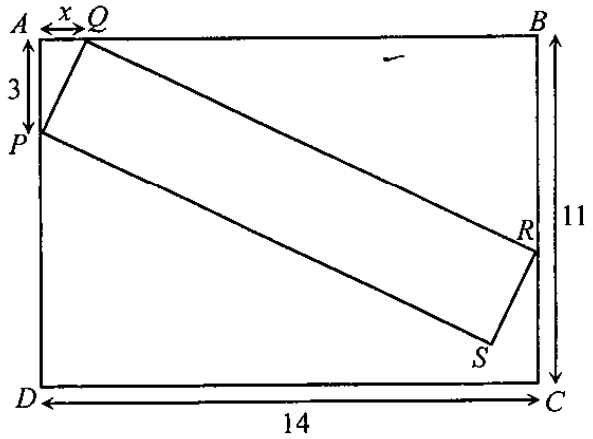


Figure 8

Figure 8 shows a rectangular cardboard *ABCD* with *BC* = 11 and *DC* = 14.

A variable rectangle, *PQRS* is cut from the cardboard according to the following rules:

[1] *P* is a fixed point on *AD* such that *AP* = 3.

[2] *Q* and *R* are points on *AB* and *BC* respectively.

Let *x* be the length of *AQ* and g (*x*) be the area of the rectangle *PQRS*.

(i) By considering ΔAPQ and ΔBQR, express *BR* in terms of *x*.

Hence show that $g(x)=\dfrac{(14-x)(x^2+9)}{3}$

(ii) By considering the fact that point *S* lies inside the cardboard *ABCD*, show that the range of values of *x* is given by $0\le x\le 2$ and $12\le x\le 14$.

(iii) Using **(a)(ii)**, find the greatest value of g(*x*) in the range in **(b)(ii).**

(7 marks)

Not for sale

More Questions in Past Papers

3. ***[A.Math 1992 Paper I Q.11]***

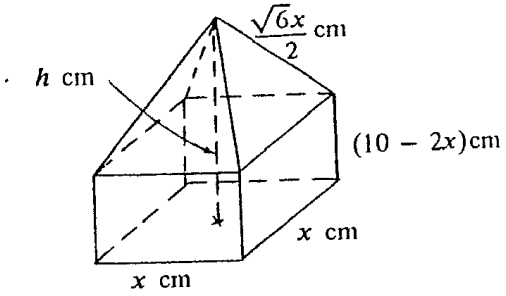


Figure 4 (a)

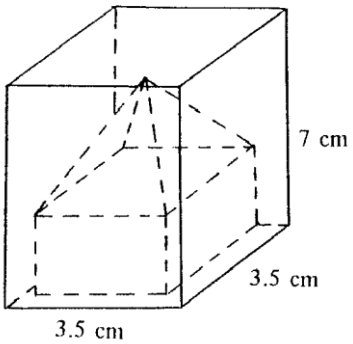


Figure 4 (b)

Figure 4(a) shows a solid consisting of a right pyramid and a cuboid with a common face which is a square of side x cm. the slant edge of the pyramid is $\frac{\sqrt{6}x}{2}$ cm and the height if the cuboid is $(10 - 2x)$ cm, where $0 < x < 5$.

(a) Let h cm be the height of the solid. Show that $h = 10 - x$. ***(3 marks)***

(b) Let V cm^3 be the volume of the solid.

(i) Show that $V = 10x^2 - \frac{5}{3}x^3$.

(ii) Find the range of values of x for which V is increasing.
Hence write down the range of values of x for which V is decreasing.
(6 marks)

(c) The solid is placed COMPLETELY inside a rectangular box as shown in Figure 4(b). The base of the box is a square of side 3.5cm and height of the box is 7cm.

(i) Show that $3 \le x \le 3.5$.

(ii) Hence find, correct to one decimal place, the greatest volume of the solid. ***(4 marks)***

(d) The side of the square base of the box in (c) is now changed to 4.7cm and the height 5.5cm. find , correct to one decimal place, the greatest volume of the solid that can be placed COMPLETELY inside the box. ***(3 marks)***



More Questions in Past Papers

4. ***[A.Math 1999 Paper I Q.13]***

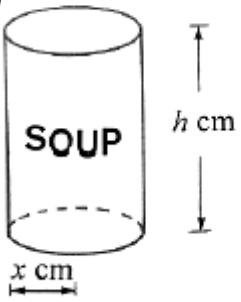


Figure 6

A food company produces cans of instant soup. Each can is in the form of a right cylinder with a base radius of x cm and a height of h cm (see Figure 6) and its capacity is V cm^3, where V is constant. The cans are made of thin metal sheets. The cost of the curved surface of the can is 1 cent per cm^2 and the cost of the plane surfaces is k cents per cm^2. Let C cents be the production cost of one can. For economic reasons, the value of C is minimized.

(a) Express h in terms of π, x and V.

Hence show that $C = \dfrac{2V}{x} + 2\pi kx^2$. ***(3 marks)***

(b) If $\dfrac{dC}{dx} = 0$, express x^3 in terms of π, k and V.

Hence show that C is a minimum when $\dfrac{x}{h} = \dfrac{1}{2k}$. ***(6 marks)***

(c) Suppose $k = 2$ and $V = 256\pi$.

(i) Find the values of x and h.

(ii) If the value of k increases, how would the dimensions of the can be affected? Explain your answer. ***(5 marks)***

(d) The company intends to produce a bigger can of capacity $2V$ cm^3, which is also in the form of a right cylinder. Suppose the costs of the curved surface and plane surfaces of the bigger can are maintained at 1 cent and k cents per cm^2 respectively. A worker suggests that the ratio of base radius to height of the bigger can should be twice that of the smaller can in order to minimize the production cost. Explain whether the worker is correct.

(2 marks)

Not for sale

More Questions in Past Papers

5. ***[A.Math 2006 Q.15]***

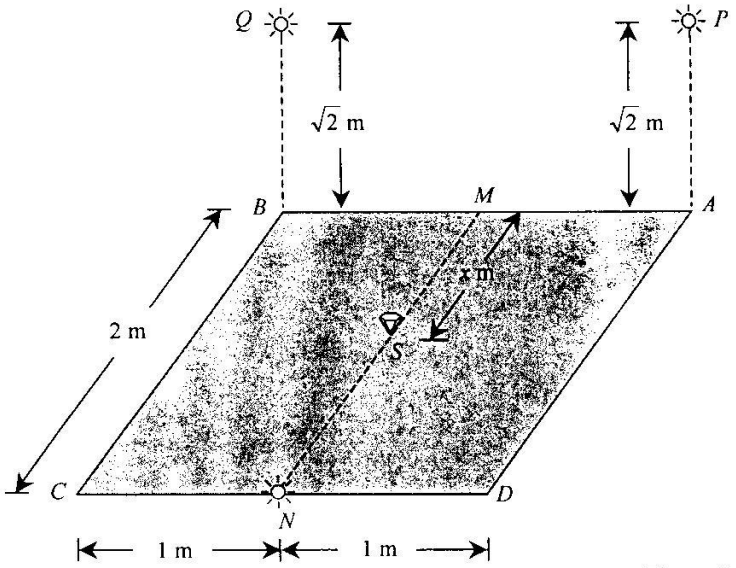


Figure 5

In Figure 5, *ABCD* is a horizontal square board of side 2 m for displaying diamonds. Let *M*, *N* be the mid-points of *BA* and *CD* respectively. Three identical small bulbs are located at points *N*, *P* and *Q* respectively for illumination purpose, where *P* and *Q* are at a height $\sqrt{2}$ m vertically above *A* and *B* respectively. A diamond is placed at a point *S* along *MN* and *MS* = *x* m, where $0 \le x \le \frac{3}{2}$. Let $PS + QS + NS = \ell$ m.

(a) Express ℓ in terms of *x*.

Hence show that $\frac{d\ell}{dx} = \frac{2x}{\sqrt{x^2+3}} - 1$. ***(2 marks)***

(b) Find the values of *x* at which ℓ attains

(i) the least value, and

(ii) the greatest value. ***(6 marks)***

(c) Suppose that the intensity of light energy received by the diamond from each bulb varies inversely as the square of the distance of the bulb from the diamond, with *k* (> 0, in suitable unit) being the variation constant. Let *E* (in suitable unit) be the total intensity of light energy received by the diamond from the three bulbs.

(i) Express *E* in terms of *k* and *x*.

(ii) A student guesses that when ℓ attains its least value, *E* will attain its greatest value. Explain whether the student's guess is correct or not. ***(4 marks)***

Not for sale

More Questions in Past Papers

6. ***[A.Math 1995 Paper I Q.12]***

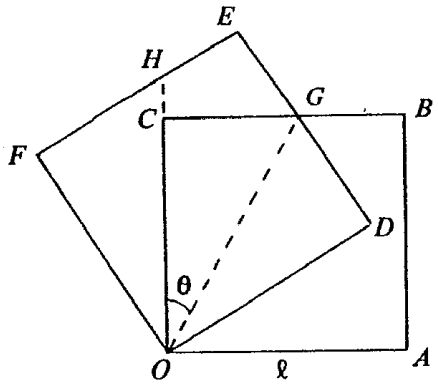


Figure 4

In Figure 4, *OABC* is the position of a square of side *l*. The square is rotated anticlockwise about *O* to a new position *ODEF*. *BC* cuts *DE* at *G* and *OC* produced cuts *EF* at *H*. Let $\angle COG = \theta$, where $\frac{\pi}{8} < \theta < \frac{\pi}{4}$.

(a) Name a triangle which is congruent to ΔOCG.

Hence show that the area of ΔOFH is $\frac{l^2}{2\tan 2\theta}$. ***(3 marks)***

(b) Let S be the sum of the areas of ΔOFH and the quadrilateral *ODGC*.

(i) Show that $S = \frac{l^2}{2}\left(\frac{2-\cos 2\theta}{\sin 2\theta}\right)$.

(ii) Find the range of values of θ for which *S* is

(1) increasing,

(2) decreasing.

Hence find the minimum value of *S*. ***(11 marks)***

(c) Find the maximum value of the area of the quadrilateral *CGEH*. ***(2 marks)***

Not for sale

More Questions in Past Papers

7 ***[A.Math 2004 Q.16]***

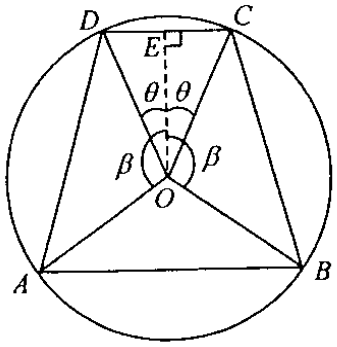


Figure 9

In Figure 9, *ABCD* is a quadrilateral inscribed in a circle centred at *O* and with radius *r*, such that *AB* // *DC* and *O* lies inside the quadrilateral.

Let $\angle COD = 2\theta$ and reflex $\angle AOB = 2\beta$, where $0 < \theta < \frac{\pi}{2} < \beta < \pi$. Point *E* denotes the foot of perpendicular from *O* to *DC*. Let *S* be the area of *ABCD*.

(a) Show that $S = \frac{r^2}{2}[\sin 2\theta - \sin 2\beta + 2\sin(\beta - \theta)]$. ***(3 marks)***

(b) Suppose β is fixed. Let S_β be the greatest value of *S* as θ varies.

Show that $S_\beta = 2r^2 \sin^3(\frac{2\beta}{3})$ and the corresponding value of θ is $\frac{\beta}{3}$.

[Hint: You may use the identity $\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$.]

(6 marks)

(c) A student says:
"Among all possible values of β, the quadrilateral *ABCD* becomes a square when S_β in (b) attains its greatest value."

Determine whether the student is correct or not. (***3 marks)***